

Permeability modulation of Fe-Ni/nanoparticle (Ni, Zn) soft magnetic composites

Sang Woo Kim, Yeon Jun Choi, Ye Rae Kim, Deok Hyeon Kim, Ye Jin In, Tae Han Kim, Bo Wha Lee^{*}

Department of Physics and Oxide Research Center, Hankuk University of Foreign Studies, Yongin, 17035, South Korea

ABSTRACT

In this study, we investigated how adding nanopowders to a micropowder affects the packing fraction and permeability. The micropowder used was a Fe-Ni crystalline alloy, which was sieved to less than 38 μm to minimize the effect of particle size distribution. The nanopowders, magnetic Ni and nonmagnetic Zn, were added to the Fe-Ni at weight ratios of 95:5, 90:10, 85:15, and 80:20. The results showed that the permeability of Fe-Ni was higher when sieved to 38 μm compared to the overall particle size. When the magnetic Ni nanopowder was added to the Fe-Ni powder sieved to 38 μm , both the packing fraction and permeability increased up to a ratio of 85:15 and then decreased. On the other hand, when the nonmagnetic Zn nanopowder was added, the packing fraction increased up to a ratio of 85:15 and then decreased, while the permeability continued to decrease.

1. Introduction

Inductors are devices that provide stable power to small battery-powered electronic devices such as smartphones. Recently, as the functions of small electronic devices have become more diverse, circuit integration and miniaturization of inductors have become necessary [1, 2]. The miniaturization of inductors can be achieved by applying soft magnetic materials with high permeability [3]. Among the soft magnetic materials, Fe-Ni alloys are used in switching regulators and noise filters due to their high saturation magnetization, DC characteristics, low coefficient of thermal expansion, and excellent temperature stability [4,5]. Enhancing the permeability of inductors can not only improve the material properties but also increase the packing fraction of the core [6].

When estimated using Ollendorff's Equation (1), the core permeability increases with not only the effective permeability of the material but also the packing fraction [7].

$$\mu = \mu_0 + \frac{\eta\mu_0(\mu_m - \mu_0)}{N(1 - \eta)(\mu_m - \mu_0) + \mu_0} \quad (1)$$

where η is the volume packing fraction, N is the coefficient of the demagnetizing field, μ_m is the permeability of the material, and μ_0 is the permeability in a vacuum. Among the methods to improve packing fraction, the bimodal method effectively enhances permeability by optimally mixing fine powders with particle sizes of less than 1 μm with micropowders [8,9]. However, the relationship between packing

fraction and permeability with the addition of magnetic and nonmagnetic powders has not been studied. Thus, this study investigated the correlation between packing fraction and permeability by adding magnetic Ni nanopowder and nonmagnetic Zn nanopowder to an Fe-Ni alloy.

2. Experimental details

2.1. Powders

The Fe-Ni powder and the Ni and Zn nanopowders used in the experiment were purchased from Avention and US-Nano, respectively. The Fe-Ni powder was sieved to less than 38 μm , and the Ni and Zn powders selected had a particle size range of 65–75 nm. The crystal structure of the powders was confirmed using an X-ray diffractometer (XRD, Rigaku MiniFlex) equipped with a Cu-K α 1 radiation source. The powder surface was confirmed using scanning electron microscopy (SEM, Jeol JSM-7600F). The atomic composition of the Fe-Ni alloy was confirmed using energy-dispersive X-ray spectroscopy (EDS; Oxford Instruments JSM-6390). The magnetic properties were confirmed using a vibrating sample magnetometer (VSM, Lakeshore VSM7404 series).

2.2. Toroidal cores

To analyze the permeability of the samples in the high-frequency

^{*} Corresponding author.

E-mail address: bwlee@hufs.ac.kr (B.W. Lee).

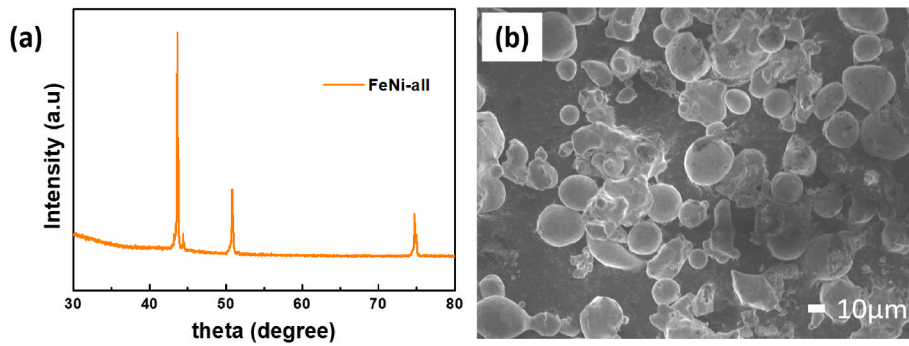


Fig. 1. (a) XRD pattern and (b) morphology of Fe-Ni powder.

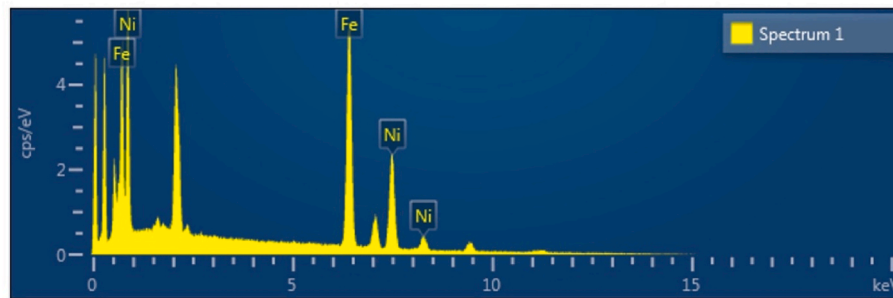


Fig. 2. EDS data of Fe-Ni powder.

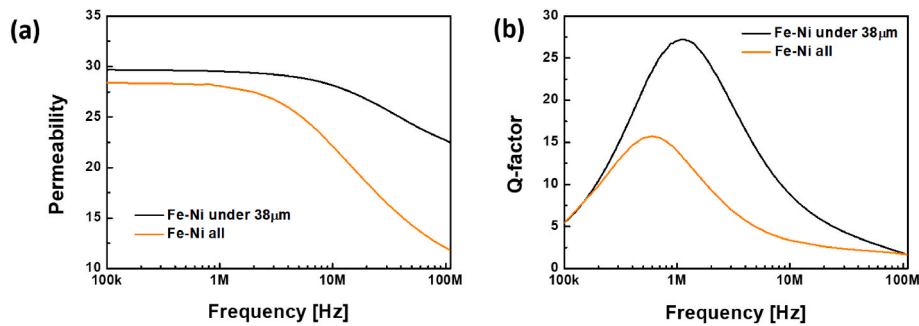


Fig. 3. (a) Permeability and (b) Q-factor of Fe-Ni powder.

range, toroidal cores were fabricated. The Fe-Ni powder consisted of the powder with the overall particle size distribution and the powder sieved to less than 38 μm . The Ni and Zn nanopowders were mixed at 5, 10, 15, and 20 wt% in Fe-Ni sieved to less than 38 μm . The core was prepared by mixing a 3 wt% epoxy binder with the powder. All samples mixed with epoxy binder were placed in a toroidal core-type mold and compressed at a pressure of 3 tons for 3 min. The compressed cores were heat treated at 160 $^{\circ}\text{C}$ for 1 h, and the outer and inner diameters of the toroidal cores were 20 mm and 13 mm, respectively. For permeability and Q-factor measurements, Cu wire was wound around the fabricated core 10 times.

The permeability and Q-factor were measured using an impedance analyzer (Hewlett Packard, 4294A) in the frequency range of 100 kHz–110 MHz.

3. Results and discussion

Fig. 1(a) shows the XRD pattern of the Fe-Ni powder between 30 and 80 $^{\circ}$. The XRD pattern shows that Fe-Ni basically has an FCC structure. However, a small BCC peak is found around 44 $^{\circ}$. This means that when alloying Fe with a BCC structure and Ni with an FCC structure, if the Fe content increases based on Fe:Ni = 4:6, both FCC and BCC structures are

present [10,11].

Fig. 1(b) shows the morphology of the Fe-Ni powder observed by SEM. The Fe-Ni powder has a near-spherical morphology, and the surface of the powder is smooth.

Fig. 2 shows the EDS results of the elemental analysis of Fe and Ni, indicating that the approximate Fe content and Ni content are 56.43 wt % and 43.54 wt%, respectively. This is consistent with the XRD results, where peaks corresponding to both BCC and FCC structures were observed.

Fig. 3 shows the permeability and Q-factor results between the frequencies of 110 kHz and 110 MHz for an un-sieved powder of Fe-Ni (Fe-Ni all) and a powder sieved to 38 μm (Fe-Ni under 38 μm). Fig. 3(a) shows the permeability according to the change in frequency. The permeability of Fe-Ni under 38 μm is higher than that of Fe-Ni all at all frequencies. This is because as the size of a powder decreases, the packing fraction increases, which increases the permeability [12]. Fig. 3 (b) shows the Q-factor used to evaluate the efficiency of the core, which is defined in Equation (2).

$$Q = \frac{\mu'}{\mu''} = \frac{1}{\tan \delta} \quad (2)$$

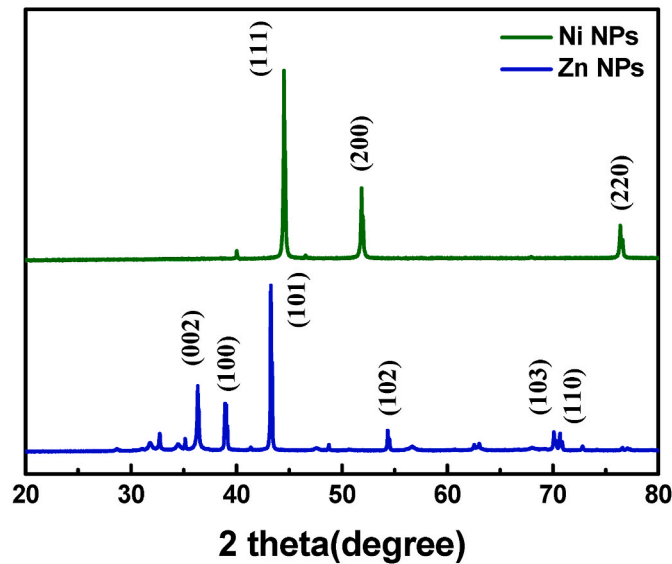


Fig. 4. XRD patterns of Ni and Zn nanopowders.

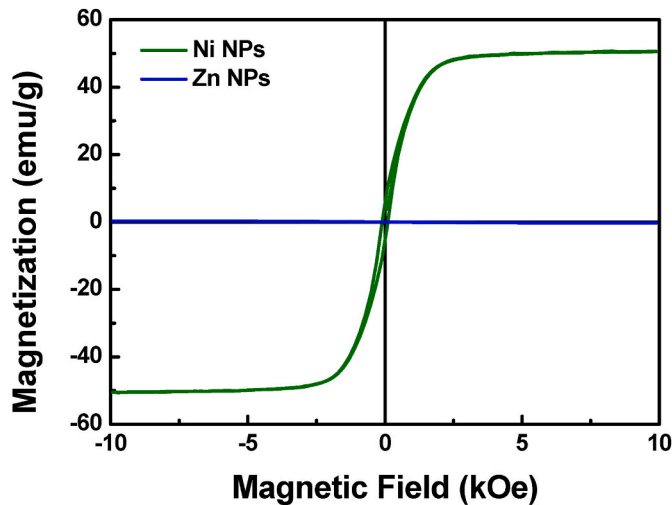


Fig. 5. Magnetic hysteresis curves of Ni and Zn nanopowders.

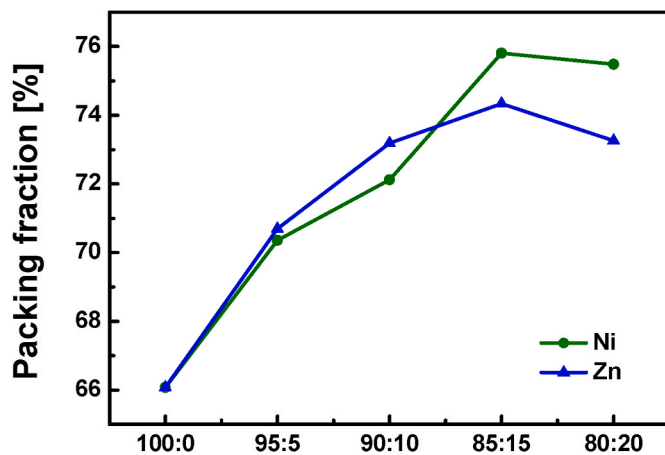


Fig. 6. Packing fraction of Fe-Ni/nanopowder composites.

Table 1

Powder density of each powder.

	Fe-Ni	Ni NPs	Zn NPs
Density (g/cm ³)	8.39	8.9	7.14

where μ' and μ'' are the real and imaginary parts of the permeability, and $\tan\delta$ is the loss tangent, which is the inverse of the Q-factor and has a direct relationship with the permeability of the core [13]. The larger the Q-factor value, the lower the loss, and for Fe-Ni, Fe-Ni under 38 μm shows a lower Q-factor value than Fe-Ni all at all frequencies, indicating that the core efficiency of the former is better than that of the latter.

Fig. 4 shows the XRD patterns of the Ni and Zn nanopowders between 30 and 80°. The Ni nanopowder has an FCC structure with (111), (200), and (220) planes, while the Zn nanopowder has a hexagonal structure with (002), (100), (101), (102), (103), and (110) planes [14,15].

Fig. 5 shows the hysteresis curves of the Ni and Zn nanopowders measured by VSM between −10 and 10 kOe. The magnetization at 10 kOe is 50.54 and −0.077 emu/g for the Ni and Zn nanopowders, respectively. The Ni nanopowder shows magnetic properties with high saturation magnetization values, while the Zn nanopowder shows negative values close to zero depending on the magnetic field. This is because Ni nanopowders are ferromagnetic materials and are magnetized in the direction of the external magnetic field, while Zn nanopowders are diamagnetic materials and are very weakly magnetized in the direction opposite to the external magnetic field [16–18].

Fig. 6 shows the variation in the packing fraction with the amount of Ni and Zn nanopowders added to Fe-Ni. Equation (3) was used to calculate the packing fraction,

$$\text{Packing fraction} = \left[\frac{d_{\text{core}}}{d_{\text{bulk}}} \times 100 \right] \% \quad (3)$$

where d_{core} and d_{bulk} are the core density and bulk density, respectively [19]. The density of each powder used in the calculation is shown in Table 1. The packing fraction continuously improves with the addition of the Ni and Zn nanopowders, increasing up to 15 wt%. This is because nanopowders fill the voids between Fe-Ni powders [9]. However, the packing fraction decreased from 20 wt%.

Fig. 7 shows the permeability results between 110 kHz and 110 MHz for the composites based on the addition of Ni and Zn nanopowders. All the permeability values were compared based on a frequency of 1 MHz. Fig. 7(a) shows how the addition of the Ni nanopowder affects the permeability. The addition of the Ni nanopowder results in higher permeability than that of pure Fe-Ni. The permeability increases up to 15 wt% but decreases from 20 wt%. Fig. 7(b) shows how the addition of the Zn nanopowder affects the permeability. The addition of the Zn nanopowder results in lower permeability than that of pure Fe-Ni, and the permeability decreases with increasing Zn content.

Fig. 8 shows the variation in the packing fraction and permeability of Fe-Ni composites with the addition of Ni and Zn nanopowders. According to Ollendorff's Equation (1), the permeability increases with the packing fraction [7]. Fig. 8(a) shows the variation in the packing fraction and permeability with the addition of the Ni nanopowder, and both increase up to 15 wt% and then decrease from 20 wt%. This indicates that the magnetic Ni nanopowder fills the voids in Fe-Ni, which improves the packing fraction and thus the permeability. The results of the addition of the Ni nanopowder are consistent with the trend of Ollendorff's Equation, confirming that the Ni nanopowder is a suitable material for improving permeability. Fig. 8(b) shows the variation in the packing fraction and permeability with the addition of the Zn nanopowder. The packing fraction increases up to 15 wt% and decreases from 20 wt%, and the permeability decreases as the amount of Zn nanopowder added increases. This means that the Zn nanopowder fills the voids in Fe-Ni, which improves the packing fraction but decreases the permeability as a nonmagnetic material. The results of the addition of

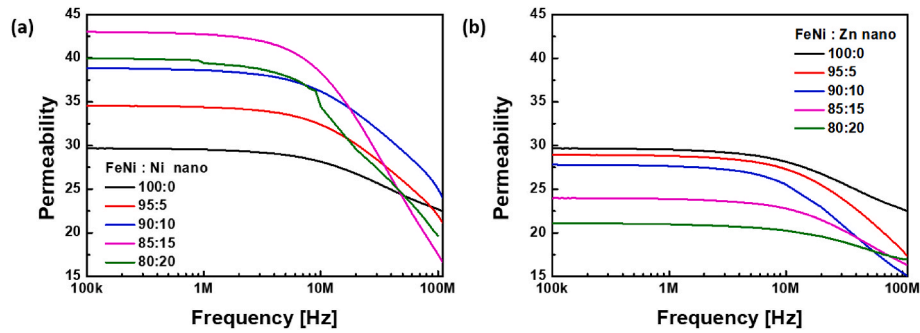


Fig. 7. Permeability of composites of (a) Fe-Ni/Ni nanopowder and (b) Fe-Ni/Zn nanopowder.

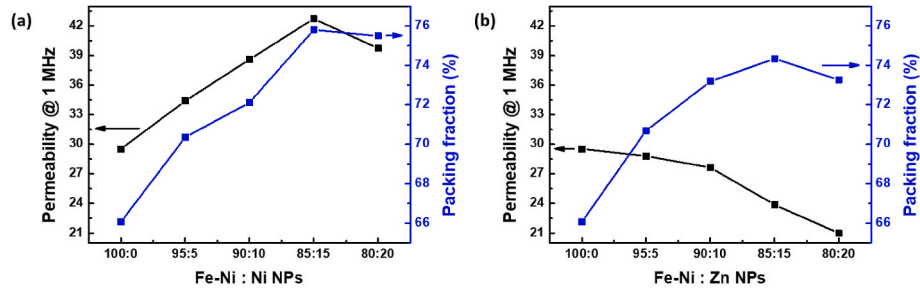


Fig. 8. Relationship between packing fraction and permeability of composites of (a) Fe-Ni/Ni nanopowder and (b) Fe-Ni/Zn nanopowder.

the Zn nanopowder are inconsistent with the trend of Ollendorff's Equation, confirming that the application of Ollendorff's Equation is limited to magnetic powders.

4. Conclusion

In this study, the electromagnetic properties of inductor cores were analyzed by adding nanopowders to Fe-Ni powder. For Fe-Ni powders, Fe-Ni under 38 μm had higher permeability and Q-factor values than Fe-Ni all. When magnetic Ni nanopowder was added to the Fe-Ni powder under 38 μm , the packing fraction and permeability increased up to 15 wt% but decreased from 20 wt%. When nonmagnetic Zn nanopowder was added, the packing fraction increased up to 15 wt% but decreased from 20 wt%. On the other hand, the permeability continuously decreased with the addition of Zn nanopowder. The results of the study confirmed that Ollendorff's Equation is restricted to magnetic powders and that the Ni nanopowder is a suitable material for improving the permeability of inductors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research was supported by the Hankuk University of Foreign Studies Research Fund of 2022.

References

- [1] Hsing-I. Hsiang, Progress in materials and processes of multilayer power inductors, *J. Mater. Sci. Mater. Electron.* 31 (2020) 16089–16110, <https://doi.org/10.1007/s10854-020-04188-8>.
- [2] Zhiyong Xia, A. Jaber, Abu Qahouq, Sushma Kotru, 3-D physical model for on-chip power inductor design with evaluation of airgap variation effect, in: 2021 IEEE Applied Power Electronics Conference and Exposition (APEC), 2021, p. 1725, <https://doi.org/10.1109/APEC42165.2021.9487062>.
- [3] Jiayi He, Yuan Han, Min Nie, Hai Guo, Hongya Yu, Zhongwu Liu c, Rong Sun, Soft magnetic materials for power inductors: state of art and future development, *Materials Today Electronics* 6 (2023) 100066, <https://doi.org/10.1016/j.mtelec.2023.100066>.
- [4] H. Shokrollahi, K. Janghorban, Soft magnetic composite materials (SMCs), *J. Mater. Process. Technol.* 189 (2007) 1–12, <https://doi.org/10.1016/j.jmatprotec.2007.02.034>.
- [5] Bo Lu, Hao Huang, Xing Long Dong, Xue Feng Zhang, Jun Peng Lei, Jian Peng Sun, Chuang Dong, Influence of alloy components on electromagnetic characteristics of core/shell-type Fe-Ni nanoparticles, *J. Appl. Phys.* 104 (2008) 114313, <https://doi.org/10.1063/1.3040006>.
- [6] Hyuk Jun Woo, Sang Woo Kim, Choongyun Paul Kim, Dong Hyeok Choi, Sumin Kim, Bo Wha Lee, Formation of Mixed-Powder composites with improved magnetic properties by adding carbonyl iron powders to FeSiCrB amorphous powder, *J. Magn. Mater.* 565 (2023) 170181, <https://doi.org/10.1016/j.jmmm.2022.170181>.
- [7] Franz Ollendorff, *Magnetostatik der Massekerne*, Arch. Elektrotechnik 25 (1931) 436–447, <https://doi.org/10.1007/BF01656937>.
- [8] Min Young Lee, Yeon Jun Choi, Seok Hee Lee, Ji Hun Ahn, Bo Wha Lee, Controlling properties of metal-polymer soft magnetic composites through microstructural deformation for power inductor applications, *J. Mater. Sci. Mater. Electron.* 33 (2022) 15763–15772, <https://doi.org/10.1007/s10854-022-08478-1>.
- [9] C.H.O.I. Yeon Jun, M.I.N. Young Lee, B.O.W.H.A. Lee, Magnetic property improvement and core-loss reduction of Fe-Si-Cr-based soft magnetic composites with addition of Fe-50Ni nanopowder, *JOM* 75 (2023) 1261–1269, <https://doi.org/10.1007/s11837-022-05650-0>.
- [10] Panpan Jing, Mengting Liu, Yongping Pu, ZhuoWang YongfeiCui, Qingfang Liu JianboWang, Dependence of phase configurations, microstructures and magnetic properties of iron-nickel (Fe-Ni) alloy nanoribbons on deoxidization temperature in hydrogen, *Sci. Rep.* 6 (2016) 37701, <https://doi.org/10.1038/srep37701>.
- [11] Wei-Su Chang, Wei Yang, Jun-Ming Guo, Feng-Jiao He, Thermal stability of Ni-Fe alloy foils continuously electrodeposited in a fluoroborate bath, *Open J. Met.* 2 (2012) 18–23, <https://doi.org/10.4236/ojmetal.2012.21003>.
- [12] Alessandro Averardi, Corrado Cola, Steven Eric Zeltmann, Nikhil Gupta, Effect of particle size distribution on the packing of powder beds: a critical discussion relevant to additive manufacturing, *Mater. Today Commun.* 24 (2020) 100964, <https://doi.org/10.1016/j.mtcomm.2020.100964>.
- [13] Qing Li, Christina W. Kartikowati, Shinji Horie, Takashi Ogi, Toru Iwaki, Kikuo Okuyama, Correlation between particle size/domain structure and magnetic properties of highly crystalline Fe₃O₄ nanoparticles, *Sci. Rep.* 7 (2017) 9894, <https://doi.org/10.1038/s41598-017-09897-5>.
- [14] Jeerapan Tientong, Stephanie Garcia, Casey R. Thurber, Teresa D. Golden, Synthesis of nickel and nickel hydroxide nanopowders by simplified chemical reduction, *Journal of Nanotechnology* (2014) 1–6, <https://doi.org/10.1155/2014/193162>.
- [15] Irfan Qasim, M. Muntaz, K. Nadeem, S. Qamar Abbas, Zinc nanoparticles at intercrystallite sites of (Cu_{0.5}Ti_{0.5})Ba₂Ca₃Cu₄O₁₂₋₈ superconductor, *J. Nanomater.* (2016) 1–6, <https://doi.org/10.1155/2016/9781790>.

- [16] Xuemin He, Wei Zhong, Chak-Tong Au, Youwei Du, Size dependance of the magnetic properties of Ni nanoparticles prepared by thermal decomposition method, *Nanoscale Res. Lett.* 8 (2013) 446, <https://doi.org/10.1186/1556-276X-8-446>.
- [17] Shashi B. Rana, R.P.P. Singh, Sandeep Arya, Structural, optical, magnetic and antibacterial study of pure and cobalt ZnO nanoparticles, *J. Mater. Sci. Mater. Electron.* 28 (2017) 2660–2672, <https://doi.org/10.1007/s10854-016-5843-0>.
- [18] Arati G. Kolhatkar, Andrew C. Jamison, Dmitri Litvinov, Richard C. Willson, T. Randall Lee, Tuning the magnetic properties of nanoparticles, *Int. J. Mol. Sci.* 14 (2013) 15977–16009, <https://doi.org/10.3390/ijms140815977>.
- [19] Pyung-Woo Jang, Bong-Han Lee, Gwang-Bo Choi, Variation of magnetic properties of Fe-Si compressed cores with Si content, *Journal of the Korean Magnetics Society* 20 (2010) 13–17, <https://doi.org/10.4283/JKMS.2010.20.1.013>.